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Inventory modeling deals with determining the level of a commodity that a business must maintain to ensure smooth operation. Inventories are vital to the successful functioning of manufacturing and retailing organisations. The inventories may consist of raw materials, work-in-progress, spare parts/consumables, and finished goods.

Inventories may be held for a variety of purpose, but in general, there are five types of inventories that an organisation can use for serving these purposes. These are

a) Movement Inventories

— Also called transit or pipeline inventories. Their existence owes to the fact that transportation time is involved in transfering substantial amounts of resources.

b) Buffer Inventories

— Are held to protect against the uncertainties of demand and supply. An organisation generally knows the average demand for various items that it needs. However, the actual demand may not exactly match the average and could well exceed it.

Note

Lead time:

Time elapsing between placing an order and having the goods in stock ready for use.

Safety stocks are held for protecting against the fluctuations in demand and lead time. The idea of keeping safety stocks is to render a higher level of customer service and consequently reduce the number of stockouts and back-order

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(c) Anticipatory Inventories

— They are held for the reason that a future demand for the product is anticipated. The idea underlying is to smoothing the production process producing for a longer duration on a continuous scale rather than operating with excessive overtime in one period and then let the system ~~to~~ be idle or close down for the reason of inadequate demand for another period.

(d) Decoupling Inventories

— the idea ~~of~~ is to decouple or disengage different parts of the production system. clearly the decoupling inventories act as shock absorbers and have cushioning effect in the face of varying user-rates, and machines breakdown and failures, and so on.

(e) Cycle Inventories

— They are held for the reason that purchase are usually made in lots rather than the exact amounts which may be needed at a point in time.
As the cycle inventories are related to the purchase in lots, they are also called lot-size inventories and the longer the lot size the greater would be the level of cycle inventories.

General Inventory (Model) Decisions

In an inventory control situation, there are three basic questions to be answered. They are:

1) How much to order?
2) When to order
3) How much ~~Safety~~ Safety stock should be kept?

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The bases for answering these questions is the minimization of the following inventory cost function:

$$\begin{pmatrix}\text{Total}\\ \text{Inventory}\\ \text{cost}\end{pmatrix} = \begin{pmatrix}\text{Purchasing}\\ \text{cost}\end{pmatrix} + \begin{pmatrix}\text{Setup}\\ \text{cost}\end{pmatrix} + \begin{pmatrix}\text{Holding}\\ \text{cost}\end{pmatrix} + \begin{pmatrix}\text{Shortage}\\ \text{cost}\end{pmatrix}$$

Inventory management systems

There are two inventory management systems. They are:

(a) Fixed order quantity system (or reorder point system)
In which a level called the re-order level is determined. As soon as the stock level reaches this point, an order for a predetermined number of units is placed.

(b) Periodic review system
where the inventories are replenished at fixed intervals of time. whereas the size of order is fixed in the previous system and the time is not, In this system the time after which the supplies are ordered is fixed but not the quantity to be ordered.

Demand and Inventory models

The demand pattern in an inventory model may assume one of four types:

1) Deterministic and constant (static) with time
2) Deterministic and variable (dynamic) with time
3) probabilistic and stationary over time
4) probabilistic and nonstationary over time.

To determine if normality is a reasonable approximation of demand is acceptable, we compute the coefficient of variation

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use it to determine the nature of demand using the following guideline

1. If the average monthly demand is "approximately" constant for all months and CV is reasonably small (< 20%), then the demand may be considered deterministic and constant with its value equal to the average of all monthly demands

2. If the average monthly demand varies appreciably among the different months but CV remains reasonably small, the demand is considered deterministic but variable.

NOVA test

3. If, in case 1, CV is high (>20%) but approximately constant, then the demand is probabilistic and stationary.

4. ~~If, in case~~ Otherwise, it is probabilistic and non-stationary.

Example

Analyse the nature of the following demand for product X

Weekly Demand

Months	~~week~~ 1	2	3	4
Jan	43	44	45	68
Feb	69	90	41	79
Mar	50	76	34	81
April	67	80	35	22
May	27	86	41	80
June	34	74	40	20

Sol^n

mean

st. dev

CoV

Use ANOVA to test for equality of means.

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	wk1	wk2	wk3	wk4	mean	std deviation	cov
Jan	43	44	45	68	50.00	10.42	20.83
Feb	69	90	41	79	69.75	18.18	26.07
Mar	50	76	34	81	60.25	19.19	31.85
Apr	67	80	39	22	52.00	22.79	43.83
May	27	86	41	80	58.50	25.08	42.88
Jun	34	74	40	20	42.00	19.85	47.26
Solution							
mean	48.33	75.00	40.00	58.33			
std deviation	15.64	14.91	3.27	26.75			
cov	32.36	19.88	8.16	45.86			

SUMMARY	*Count*	*Sum*	*Average*	*Variance*
Jan	4	200	50	144.6667
Feb	4	279	69.75	440.9167
Mar	4	241	60.25	490.9167
Apr	4	208	52	692.6667
May	4	234	58.5	839
Jun	4	168	42	525.3333
wk1	6	290	48.33333	293.4667
wk2	6	450	75	266.8
wk3	6	240	40	12.8
wk4	6	350	58.33333	858.6667

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Months	1837.333	5	367.4667	1.035831	0.432306	2.901295
Weeks	4079.167	3	1359.722	3.832843	0.032085	3.287382
Error	5321.333	15	354.7556			
Total	11237.83	23				

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Inventory Costs

In determining an optimal inventory policy, all or some of these cost components are included in the objective function

purchase cost

This refers to the nominal cost of inventory. it is the purchase price for the items that are sought from outside sources and the production cost of items are produced within the organisation.

ii ordering cost / set-up cost

This is incurred whenever the inventory is replenished. it includes costs associated with the processing and chasing of the purchase order, transportation, inspection for quality, expediting overdue order etc it is also known as procurement cost.

iii Carrying cost

Also known as holding or storage cost, carrying cost represents the cost that is associated with the purchasing and storing an item in inventory. its elements include the opportunity cost of capital of is invested in the stock; the costs directly associated with storing goods; the obsolescence cost; deterioration costs and cost incurred in preventing deteriorations, and fire and general insurance etc.

(iv) Stockout cost

The cost associated with not serving the customers. stockouts imply shortages. this could be internal or external.

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Fixed-order Quantity Systems

The classical EOQ Model (Wilson formulation)

EOQ ---- Economic order quantity model

Under the EOQ model, a cost model is first developed, and then it is manipulated to form an inventory model.

Assumptions

i. The demand for the item is certain, continuous and constant with time

ii. the lead-time, that is, the time between placing an order and its delivery is known and fixed. Thus when the lead time is zero, the delivery is instantaneous.

iii. within the range of the quantities to be ordered, the per unit holding cost and the ordering cost are constant and independent of the quantity ordered.

iv. the purchase price of the item is constant, no discount is available on purchase of large lots.

v. the inventory is replenished immediately as the stock level reaches zero. Consequently, there are no stock overages or shortages

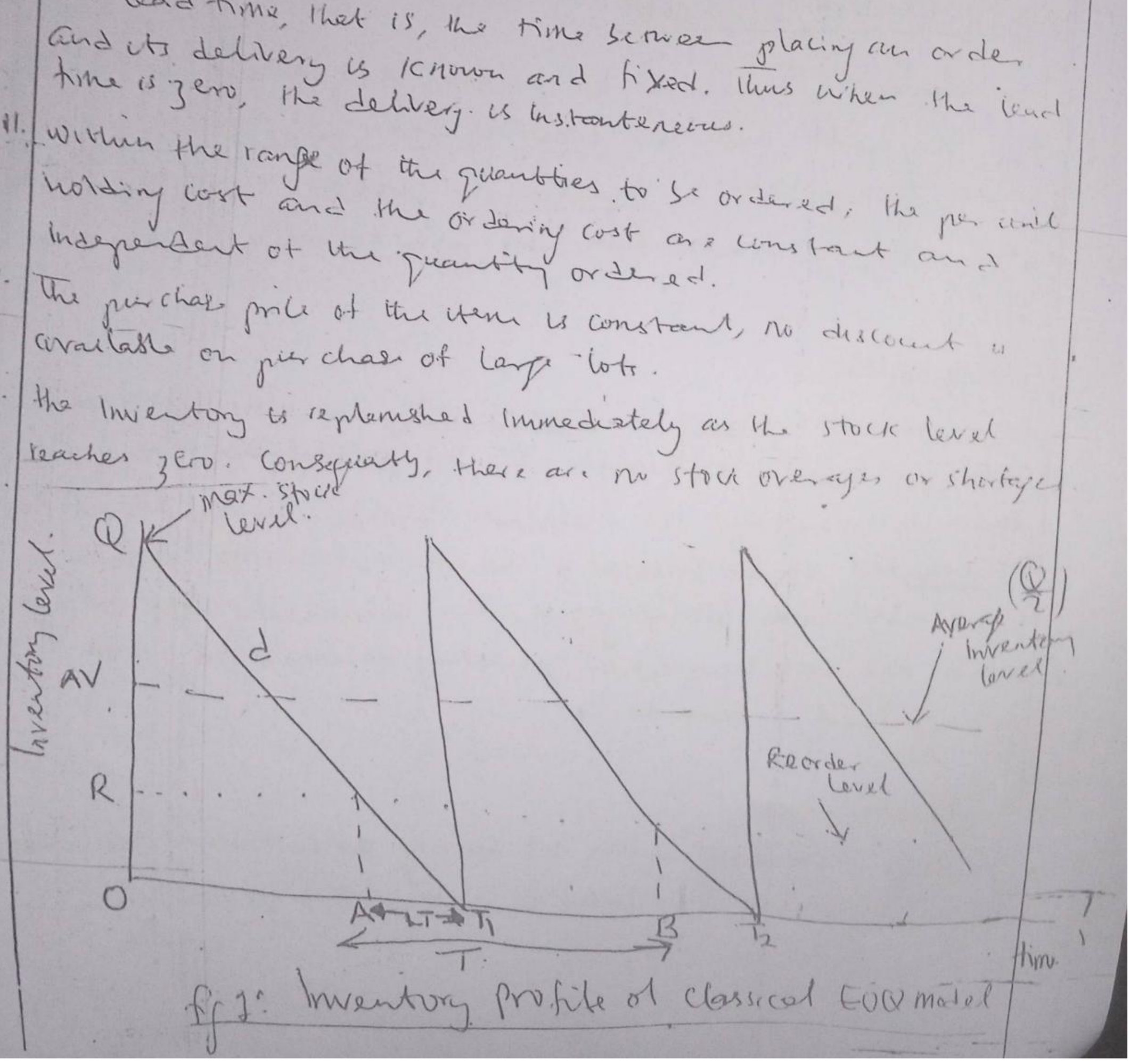


fig 1: Inventory profile of classical EOQ model

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The cost model is

$$T(Q) = O(Q) + H(Q)$$

where

Q — the ordering quantity

$T(Q)$ — total (variable) annual inventory cost

$O(Q)$ — total annual ordering cost

$H(Q)$ — total annual holding cost.

Example

Ade Ltd produces 2000 TV sets in a year for which it needs an equal number of tubes of a certain type. Each tube costs ₦10 ('000) and the cost of to hold a tube in stock for a year is ₦2.40. Besides, the cost of placing an order is ₦150, which is not related to its size.

Now, if an order for 2,000 tubes is placed, only one order per annum is required. When 1000 units are ordered for, 2 orders in a year are needed, while if 500 units are ordered to be supplied, then a total of 4 orders per annum are required. Naturally, as the number of orders placed increases the ordering cost goes up. More orders, however, would also imply smaller order quantity and, therefore, decreasing holding costs. Thus we have have a trade-off between the ordering and the holding cost.

find the order size that minimizes the cost function $T(Q)$.

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Sol:

(a) Total annual ordering cost

This is given by the number of times an order is placed, multiplied by the ordering cost per order, A.

$$O(Q) = N \times A$$

Where $N = \frac{D}{Q} = \frac{\text{annual demand}}{\text{order quantity}}$

$$O(Q) = \frac{D}{Q} \cdot A$$

For the example when

$N = 1$, $Q = 2000$ and $O(Q) = 1 \times 150 = 150$

$N = 2$, $Q = 1000$ and $O(Q) = 2 \times 150 = 300$

$N = 4$, $Q = 500$ and $O(Q) = 4 \times 150 = 600$

$N = 5$, $Q = 400$ and $O(Q) = 5 \times 150 = 750$

(b) Total annual holding cost:

Obtained by multiplying the unit holding cost, h, by the average number of units held in the inventory, $Q/2$

$$H(Q) = \frac{Q}{2} \cdot h$$

For our example, when

$Q = 2000$, $H(Q) = \frac{2000}{2} \times 2.40 = 2400$

$Q = 1000$, $H(Q) = \frac{1000}{2} \times 2.40 = 1200$

$Q = 500$, $H(Q) = \frac{500}{2} \times 2.40 = 600$

$Q = 400$, $H(Q) = \frac{400}{2} \times 2.40 = 480$

Note

If i is the proportion (or percentage), which is also known as the holding rate, and c is the cost of the item, the unit holding cost denoted h is

$$h = ic$$

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(c) Total annual Inventory cost (Variable) obtained by adding VR both components of (a) and (b).

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(a) Calculation of Total cost

Order quantity	Annual ordering cost $O(Q)$	Annual Holding cost $H(Q)$	Total Annual cost $T(Q) = O(Q) + H(Q)$
100	3000	120	3120
200	1500	240	1740
400	750	480	1230
500	600	600	1200
1000	300	1200	1500
2000	150	2400	2550
	$\frac{2000}{Q} \times 150$	$\frac{Q}{2} \times 240$	

Determination of the re-order level

Suppose the following additional data are given

No of working days in the year = 250

~~Lott~~ Lead time = 15 working days

With this information, the daily demand = 2000/250 = 8

Demand during lead time = 15 × 8 = 120

∴ Re-order level = 120

Annual total variable inventory cost

The minimum annual Inventory cost can be determined by substituting Q^* for Q in the cost equation, thus

$$T(Q^*) = \frac{D}{Q^*} A + \frac{Q^*}{2} h$$

$$= \frac{DA}{\sqrt{\frac{2AD}{h}}} + \frac{\left(\sqrt{\frac{2AD}{h}}\right) h}{2}$$

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$\Rightarrow$

$$T(Q^*) = \sqrt{2ADh}$$

∴ for the example

$$T(Q^*) = \sqrt{2 \times 150 \times 2000 \times 2.40}$$

$$= \sqrt{1440000} \quad = 1200$$

Example

Using the following data, obtain the EOQ and the total variable cost associated with the policy of ordering quantities of the size:

Annual Demand = 20,000

ordering cost = 150 per order

Inventory carrying cost = 24% of average Inventory value

Soln

C D = ₦20,000 , C — unit price

A = 150/order D — demand

i = 24% = 0.24

EOQ $= \sqrt{\frac{2 \times 150 \times 20{,}000}{0.24}}$

$= 5000$

Total cost $T(Q^*) = \sqrt{2 \times 150 \times 24{,}000 \times 0.24} = 1200$

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A — Annual ordering cost

D — Annual Demand

h — holding cost per unit

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$$EOQ = \sqrt{\frac{2ACD}{i}}$$

Total cost

$$T(Q^*) = \sqrt{2ACDi}$$

CD —— the annual demand

A —— the acquisition cost

i —— the holding rate.

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